

BHUVISION // SatQuery AI

Enterprise Production Deployment & Systems Engineering Manual

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1. System Architecture & Core Stack

BHUVISION (SatQuery AI) is an autonomous multi-sensor satellite Earth intelligence platform designed to ingest, fuse, and analyze multi-spectral optical imagery (Sentinel-2, Cartosat-3, Landsat-9) and Synthetic Aperture Radar (SAR Sentinel-1, RISAT-1B). The system features an autonomous 9-agent reasoning graph, photorealistic 3D Google Earth terrain elevation, authenticated NASA GIBS and MapTiler feeds, Google Maps Platform hybrid vector rendering, and a bank-grade multi-factor authentication suite.

Layer	Component Technology	Operational Function
Backend REST API	FastAPI (Python 3.13) + Uvicorn + Pydantic v2	Async high-throughput endpoints, pipeline execution, telemetry, proxying
3D Google Earth	MapLibre GL JS 3.6.2 + AWS Terrarium DEM + Three.js r128	60 FPS GPU 3D terrain elevation (DEM), 3D building extrusions, Rayleigh sky haze
Geospatial Engine	Google Maps + MapTiler + NASA GIBS + TopoJSON	0.15m Google Hybrid, 0.5m MapTiler, NRT 250m NASA GIBS, TopoJSON boundaries
AI Agent Council	OpenRouter Gateway (9 LLM Specialist Models)	Gemini 2.5/2.0, Claude 3.7 Sonnet, Qwen 2.5 72B, DeepSeek R1, Llama 3.3 70B
Authentication	Google SSO (OAuth 2.0) + Email/Pass + Phone SMS OTP	JWT Bearer tokens (24h TTL), bcrypt hashing, 6-digit cryptographic OTP
Mobile PWA	Web App Manifest + Standalone Mobile Service Worker	Installable standalone mobile application for Android, iOS, and Desktop

2. Multi-Factor Authentication Setup

BHUVISION implements a defense-in-depth authentication suite supporting three complementary workflows: Google Single Sign-On (OAuth 2.0), Email/Password Registration, and Phone SMS OTP Verification.

2.1 Google OAuth 2.0 Setup (Step-by-Step Google Cloud Console)

1. Navigate to the **Google Cloud Console** (console.cloud.google.com) and create or select your project (e.g. `bhuvision-prod`).

2. Navigate to **APIs & Services** → **OAuth consent screen**:

- Select **External** User Type and click *Create*.
- Set App name: `BHUVISION Space Intelligence`.
- Set User support email and Developer contact information.
- Under **Scopes**, add `openid`, `.../auth/userinfo.email`, and `.../auth/userinfo.profile`.

3. Navigate to **APIs & Services** → **Credentials** → click **Create Credentials** → **OAuth Client ID**:

- Application type: **Web application**.
- Name: `BHUVISION Web Client`.
- **Authorized JavaScript origins**:

```
http://localhost:8000
http://127.0.0.1:8000
https://bhuvision.yourdomain.com
```

- **Authorized redirect URIs**:

```
https://bhuvision.yourdomain.com/api/auth/google/callback
```

4. Click **Create**. Copy your `Client ID` and `Client Secret` into your `.env` file:

```
GOOGLE_CLIENT_ID=your-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=60CSPX-your-client-secret
```

2.2 Email & Password JWT Clearance

Analysts can register with an organizational email, secure password, and Security Clearance Level (Level 1 Observer, Level 2 Geospatial Analyst, Level 3 Mission Commander).

- **Password Security:** Passwords are salt-hashed using `bcrypt` with a minimum work factor of 12 rounds.
- **JWT Tokens:** Issued upon successful registration or login with a 24-hour expiration (86,400 seconds) signed with `HS256` or `RS256`.
- **Endpoints:**
 - `POST /api/auth/register`: Registers new analyst account.
 - `POST /api/auth/login`: Authenticates credentials and returns JWT Bearer token.
 - `GET /api/auth/me`: Retrieves active user clearance profile using `Authorization: Bearer <token>`.

2.3 Phone SMS OTP Verification Setup

The system generates a cryptographically random 6-digit numeric OTP (e.g. `849201`) with a 300-second (5 minute) Time-to-Live (TTL).

For local testing and offline hackathon demonstration, the platform provides secure simulated delivery with demo hint bypass (or master judge bypass code `123456` on `+919999999999`). For live production dispatch, configure your chosen SMS gateway in `.env` :

```
# Choose gateway: twilio | aws_sns | fast2sms | msg91
SMS_GATEWAY_PROVIDER=twilio

# Twilio Configuration
TWILIO_ACCOUNT_SID=ACXXXXXXXXXXXXXXXXXXXXXXXXXXXX
TWILIO_AUTH_TOKEN=your_twilio_auth_token
TWILIO_PHONE_NUMBER=+1234567890

# Fast2SMS Configuration (India Domestic)
FAST2SMS_API_KEY=your_fast2sms_api_key

# AWS SNS SMS Configuration
AWS_ACCESS_KEY_ID=AKIAXXXXXXXXXXXXXXXX
AWS_SECRET_ACCESS_KEY=your_aws_secret_access_key
AWS_DEFAULT_REGION=ap-south-1
```

3. Geospatial & Map APIs Configuration

BHUVISION integrates three commercial and scientific map APIs concurrently to provide complete coverage, zero-failure resilience, and sub-meter ground sampling distance (GSD).

3.1 Google Maps Platform (Vector 3D Hybrid)

- **Features:** 0.15m–0.3m GSD commercial satellite imagery, vector photogrammetry, 3D tilt up to 67.5°, and global geocoding.
- **Setup:**
 1. Go to [Google Cloud Maps Platform](#).
 2. Enable **Maps JavaScript API** and **Geocoding API**.
 3. Create an API Key and restrict it by **HTTP referrers** (e.g. `*.yourdomain.com/*`).
 4. In GCP Console under *Map Management*, create a **Map ID** with *Map type: JavaScript Vector* and *Tilt: Enabled*.
 5. Configure key in `.env` :

```
GOOGLE_MAPS_API_KEY=AIzaSyA3Dj5LIpk0mUF4e7kUmKYpvNZNkfWyniU
```

3.2 MapTiler Cloud (3D Terrain-RGB & Satellite)

- **Features:** 0.5m GSD global satellite mosaic and Terrain-RGB elevation tiles for true 3D Digital Elevation Models.
- **Setup:**
 1. Sign up at [cloud.maptiler.com](#).
 2. Go to **Account** → **Keys** and copy your key.
 3. Under key settings, configure Allowed Domains (e.g. `localhost:8000`, `yourdomain.com`).
 4. Configure key in `.env` :

```
MAPTILER_API_KEY=229a5cc6-0f58-4122-beab-fcc3948cf451
```

3.3 NASA Earthdata GIBS (Authenticated Bearer JWT Proxy)

- **Features:** Near-Real-Time (NRT) 250m resolution daily Earth reflectance from MODIS Terra/Aqua and VIIRS.
- **Setup:**
 1. Register an account on [NASA Earthdata Login \(URS\)](#).
 2. Under *Authorized Apps*, approve *GIBS API / Global Imagery Browse Services*.
 3. Generate a long-lived Bearer JWT token from your NASA Earthdata profile.
 4. Configure the token in `.env` :

```
NASA_EARTHDATA_TOKEN=eyJ0eXAiOiJKV1QiLCJvcmlnaW4iOiJFYXJ0aGRhdGEgTG9naW4i...
NASA_GIBS_ENDPOINT=https://gibs.earthdata.nasa.gov/wmts/epsg3857/best
```

5. BHUVISION automatically proxies GIBS tiles through `/api/nasa-tile`, injecting the Bearer Authorization header to prevent browser CORS restrictions.

3.4 AWS Open Data Terrarium DEM (Zero-Key 3D Fallback)

- **Features:** Free, open-access, global 3D Digital Elevation Model tiles hosted on Amazon Web Services Open Data.
- **Tile URL:** `https://s3.amazonaws.com/elevation-tiles-prod/terrarium/{z}/{x}/{y}.png`
- **Elevation Formula:** $\text{Elevation (m)} = (R * 256 + G + B / 256) - 32768$
- **Integration:** Used by MapLibre GL JS as a zero-key fallback source, guaranteeing that 3D Google Earth terrain never fails even if commercial API quotas are exceeded.

4. 9-Agent OpenRouter LLM Gateway Setup

BHUVISION decomposes user questions into a multi-agent directed acyclic graph (DAG) executed across 9 specialized LLM models connected via the OpenRouter gateway.

Agent Role	Configured LLM Model	Reasoning Specialty
Lead Orchestrator	google/gemini-2.5-pro	High-level task decomposition, agent consensus, synthesis
Query Planner	anthropic/claude-3.7-sonnet	Temporal bounding, sensor requirement definition
Sensor Router	meta-llama/llama-3.3-70b-instruct	Spectral band routing (Optical vs SAR vs NDVI)
SAR Radar Expert	deepseek/deepseek-r1	Dielectric roughness, double-bounce, flood backscatter
Change Detection	qwen/qwen-2.5-72b-instruct	Bi-temporal differencing and morphological analysis
Visual Grounding	google/gemini-2.0-flash-exp	Bounding polygon coordinates and MGRS projection
VQA Analyst	anthropic/claude-3.5-sonnet	Visual Question Answering on high-res tiles
Evidence Fusion	mistralai/mistral-large-2407	Multi-sensor synthesis and confidence calibration
Audit & Trace	openai/gpt-4o-mini	Immutable trace ledger and chain-of-custody

```
# OpenRouter Configuration in .env
OPENROUTER_API_KEY=your_openrouter_api_key_here
OPENROUTER_BASE_URL=https://openrouter.ai/api/v1
MODEL_SERVING=gateway
DEFAULT_ORCHESTRATOR_MODEL=google/gemini-2.5-pro
```

5. Production Deployment Workflows

Option A: Docker & Docker Compose (Recommended)

The containerized deployment packages the FastAPI backend, Uvicorn ASGI server, and static assets in a multi-stage Alpine/Debian image.

```
# 1. Clone repository to deployment server
git clone https://github.com/Ayushnot41/Satquery-Ai.git /opt/bhuvision
cd /opt/bhuvision

# 2. Copy and configure environment secrets
cp .env.example .env
nano .env

# 3. Build and launch containerized services in background
docker compose up -d --build

# 4. Verify running container status and logs
docker compose ps
docker compose logs -f backend

# 5. Check API health
curl -s http://127.0.0.1:8000/api/health | jq
```

Option B: Ubuntu 24.04 LTS Cloud VM (AWS EC2 / DigitalOcean / GCP)

Deploy directly on a Linux virtual machine using a systemd service daemon behind an Nginx reverse proxy.

```
# 1. Update system packages & install dependencies
sudo apt-get update && sudo apt-get install -y python3.13 python3.13-venv nginx certbot python3-certbot-nginx git curl

# 2. Set up application directory & virtual environment
sudo mkdir -p /opt/bhuvision
sudo chown -R $USER:$USER /opt/bhuvision
git clone https://github.com/Ayushnot41/Satquery-Ai.git /opt/bhuvision
cd /opt/bhuvision

python3.13 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r backend/requirements.txt

# 3. Create Systemd Service File
sudo tee /etc/systemd/system/bhuvision.service << 'EOF'
[Unit]
Description=BHUVISION SatQuery AI Production Service
After=network.target

[Service]
User=www-data
Group=www-data
WorkingDirectory=/opt/bhuvision/backend
EnvironmentFile=/opt/bhuvision/.env
ExecStart=/opt/bhuvision/venv/bin/uvicorn app.main:app --host 127.0.0.1 --port 8000 --workers 4 --proxy-headers
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target
EOF

# 4. Start and enable systemd daemon
sudo systemctl daemon-reload
sudo systemctl start bhuvision
sudo systemctl enable bhuvision
sudo systemctl status bhuvision
```

Option C: Managed Cloud Container (Google Cloud Run / AWS App Runner / Render)

```
# Deploy to Google Cloud Run with gcloud CLI
gcloud run deploy bhuvision --source . --platform managed --region asia-south1 --allow-unauthenticated --port 8000 --set-env-vars ENVIRONMENT=production,DEBUG=false
```

Option D: On-Premise Windows Server (NSSM Windows Service)

```
# 1. Download NSSM (Non-Sucking Service Manager)
# 2. Open Administrator PowerShell and install service:
nssm install BhuvisionService "C:\Program Files\Python313\python.exe" "-m uvicorn app.main:app --host 0.0.0.0 --port 8000 --workers 4"
nssm set BhuvisionService AppDirectory "G:\Satquery Ai\backend"
nssm set BhuvisionService Start SERVICE_AUTO_START
nssm start BhuvisionService

# 3. Add Windows Firewall Rule for Port 8000
New-NetFirewallRule -DisplayName "BHUVISION Web Port" -Direction Inbound -LocalPort 8000 -Protocol TCP -Action Allow
```

6. Nginx Reverse Proxy & SSL Configuration

Configure Nginx to terminate SSL/TLS, compress static assets with Gzip, support WebSockets, and proxy API traffic to Uvicorn.

```
# /etc/nginx/sites-available/bhuvision
server {
    listen 80;
    server_name bhuvision.yourdomain.com;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name bhuvision.yourdomain.com;

    ssl_certificate /etc/letsencrypt/live/bhuvision.yourdomain.com/fullchain.pem;
    ssl_certificate_key /etc/letsencrypt/live/bhuvision.yourdomain.com/privkey.pem;
    ssl_protocols TLSv1.2 TLSv1.3;
    ssl_ciphers HIGH:!aNULL:!MD5;

    # Gzip Compression for 3D SVGs and GeoJSON
    gzip on;
    gzip_vary on;
    gzip_min_length 1024;
    gzip_types text/plain text/css application/json application/javascript text/xml image/svg+xml;

    # Max upload size for satellite imagery
    client_max_body_size 64M;

    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_http_version 1.1;
        proxy_set_header Upgrade $http_upgrade;
        proxy_set_header Connection "upgrade";
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 180s;
    }
}
```

```
# Enable site and issue free SSL certificate
sudo ln -s /etc/nginx/sites-available/bhuvision /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl reload nginx
sudo certbot --nginx -d bhuvision.yourdomain.com --non-interactive --agree-tos -m admin@yourdomain.com
```

7. Mobile Web App (PWA & APK Packaging)

BHUVISION is fully engineered as a Progressive Web App (PWA) with a manifest hosted at `/manifest.json`.

- **Android Mobile Chrome Installation:** Open `https://bhuvision.yourdomain.com/app` → Tap browser three dots → Tap **"Install App"** or **"Add to Home Screen"**. Launches as a fullscreen native app without URL bars.
- **iOS Mobile Safari Installation:** Open in Safari → Tap the **Share** button (box with up arrow) → Tap **"Add to Home Screen"** → Tap *Add*.
- **Google Play Store APK Packaging:**

```
# Generate signed Android APK using Bubblewrap CLI
npm install -g @bubblewrap/cli
bubblewrap init --manifest="https://bhuvision.yourdomain.com/manifest.json"
bubblewrap build
```

Alternatively, paste your URL at [PWABuilder.com](https://pwabuilder.com) to download signed packages for Google Play Store, Microsoft Store, and Apple App Store.

8. Complete Production Environment Template (.env.production)

```
# =====
# BHUVISION PRODUCTION ENVIRONMENT CONFIGURATION
# =====
ENVIRONMENT=production
DEBUG=false
APP_NAME=BHUVISION
APP_VERSION=1.0.0
ALLOWED_HOSTS=bhuvision.yourdomain.com,localhost,127.0.0.1

# Security & JWT Tokens
JWT_SECRET_KEY=generate_with_openssl_rand_hex_32_high_entropy_secret
JWT_ALGORITHM=HS256
ACCESS_TOKEN_EXPIRE_MINUTES=1440

# Google OAuth 2.0 Web Client
GOOGLE_CLIENT_ID=your-google-client-id.apps.googleusercontent.com
GOOGLE_CLIENT_SECRET=GOCSPX-your-google-client-secret
GOOGLE_REDIRECT_URI=https://bhuvision.yourdomain.com/api/auth/google/callback

# SMS Gateway (Twilio / AWS SNS / Fast2SMS)
SMS_GATEWAY_PROVIDER=twilio
TWILIO_ACCOUNT_SID=ACXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX
TWILIO_AUTH_TOKEN=your_twilio_auth_token
TWILIO_PHONE_NUMBER=+1234567890

# Three Commercial & Scientific Map APIs
GOOGLE_MAPS_API_KEY=AIzaSyA3Dj5LIpk0mUF4e7kUmKYpvNZNkfWyniU
MAPTILER_API_KEY=229a5cc6-0f58-4122-beab-fcc3948cf451
NASA_EARTHDATA_TOKEN=eyJ0eXAiOiJKV1QiLCJvcmlnaW4iOiJFYXJ0aGRhdGEgTG9naW4i...
NASA_GIBS_ENDPOINT=https://gibs.earthdata.nasa.gov/wmts/epsg3857/best

# OpenRouter 9-Agent AI Gateway
OPENROUTER_API_KEY=your_openrouter_api_key_here
OPENROUTER_BASE_URL=https://openrouter.ai/api/v1
MODEL_SERVING=gateway
DEFAULT_ORCHESTRATOR_MODEL=google/gemini-2.5-pro
```

9. Verification, Testing & Runbook

```
# 1. Run full 44-test backend pytest suite
pytest backend/tests/ -v --tb=short

# 2. Verify all 3D SVG assets and XML well-formedness
pytest backend/tests/test_svg_assets_validity.py -v

# 3. Test PDF Download endpoint
curl -I http://127.0.0.1:8000/api/download/deployment-manual

# 4. Check backend health probe
curl -s http://127.0.0.1:8000/api/health | jq
```

Production Disaster Recovery & Common Error Codes:

- **HTTP 401 Unauthorized:** JWT token expired or Bearer header missing. Analysts must re-authenticate at `/api/auth/login`.
- **HTTP 403 Forbidden on Tiles:** Check MapTiler or Google Cloud HTTP referrer restrictions. Add your live domain.
- **HTTP 502 Bad Gateway:** Uvicorn daemon down. Run `sudo systemctl restart bhuvision` and inspect logs with `journalctl -u bhuvision -e`.